Lesson 18: Higher Order Linear Nonhomogeneous Differential Equations with Constant Coefficients

Question 1:

Find the general solution of the linear nonhomogeneous differential equation $2y''' - 5y'' - 11y' - 4y = x^2 + 2x - 7$

The linear homogeneous differential equation is $2y''' - 5y'' - 11y' - 4y = 0$.

The indicial equation is $2r^3 - 5r^2 - 11r - 4 = 0$.

Note: $2r^3 - 5r^2 - 11r - 4 = (2r + 1)(r + 1)(r - 4)$.

a) The roots of the indicial equation $2r^3 - 5r^2 - 11r - 4 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, and $r_3 =$ ___?___.

b) The complementary solution $y_c(x)$ of $2y''' - 5y'' - 11y' - 4y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_p(x)$ for $g(x) = x^2 + 2x - 7$ is $y_p(x) =$ ___?___.

d) The general solution of the linear nonhomogeneous differential equation $2y''' - 5y'' - 11y' - 4y = x^2 + 2x - 7$ is $y(x) = y_c(x) + y_p(x) =$ ___?___.

Question 2:

Find the general solution of the linear nonhomogeneous differential equation $4y''' - 8y'' - 51y' - 44y = \sin(x) + x^2$

The linear homogeneous differential equation is $4y''' - 8y'' - 51y' - 44y = 0$.

The indicial equation is $4r^3 - 8r^2 - 51r - 45 = 0$.

Note: $4r^3 - 8r^2 - 51r - 45 = (2r+3)^2(r-5)$.

a) The roots of the indicial equation $4r^3 - 8r^2 - 51r - 45 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, and $r_3 =$ ___?___.
Note: $r_1 = r_2$ is a repeated root.

b) The complementary solution $y_c(x)$ of $4y''' - 8y'' - 51y' - 44y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_{p1}(x)$ for $g_1(x) = \sin(x)$ is $y_{p1}(x) =$ ___?___.

d) The particular solution $y_{p2}(x)$ for $g_2(x) = x^2$ is $y_{p2}(x) =$ ___?___.

e) The general solution of the linear nonhomogeneous differential equation $4y''' - 8y'' - 51y' - 44y = \sin(x) + x^2$ is $y(x) = y_c(x) + y_{p1}(x) + y_{p2}(x) =$ ___?___.

Question 3:

Find the general solution of the linear nonhomogeneous differential equation $2y''' - y'' - 25y' - 12y = \sin(x) + \cos(x)$

The linear homogeneous differential equation is $2y''' - y'' - 25y' - 12y = 0$.

The indicial equation is $2r^3 - r^2 - 25r - 12 = 0$.

Note: $2r^3 - r^2 - 25r - 12 = (r+3)(r-4)(2r+1)$.

a) The roots of the indicial equation $2r^3 - r^2 - 25r - 12 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, and $r_3 =$ ___?___.

b) The complementary solution $y_c(x)$ of $2y''' - y'' - 25y' - 12y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_p(x)$ for $g(x) = \sin(x) + \cos(x)$ is $y_p(x) =$ ___?___.

Hint: $y_p(x) = A \cdot \sin x + B \cdot \cos x$; $y_p'(x) = A \cdot \cos x - B \cdot \sin x$; $y_p''(x) = -A \cdot \sin x - B \cdot \cos x$; and $y_p'''(x) = -A \cdot \cos x + B \cdot \sin x$

d) The general solution of the linear nonhomogeneous differential equation $2y''' - y'' - 25y' - 12y = \sin(x) + \cos(x)$ is $y(x) = y_c(x) + y_p(x) =$ ___?___.

Question 4:

Find the general solution of the linear nonhomogeneous differential equation $2y''' - 3y'' - 11y' + 6y = xe^{2x}$

The linear homogeneous differential equation is $2y''' - 3y'' - 11y' + 6y = 0$.

The indicial equation is $2r^3 - 3r^2 - 11r + 6 = 0$.

Note: $2r^3 - 3r^2 - 11r + 6 = (r+2)(r-3)(2r-1)$.

a) The roots of the indicial equation $2r^3 - 3r^2 - 11r + 6 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, and $r_3 =$ ___?___.

b) The complementary solution $y_c(x)$ of $2y''' - 3y'' - 11y' + 6y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_p(x)$ for $g(x) = xe^{2x}$ is $y_p(x) =$ ___?___.

Hint: $y_p(x) = (Ax + B)e^{2x}$; $y_p'(x) = (2Ax + A + 2B)e^{2x}$; $y_p''(x) = (4Ax + 4A + 4B)e^{2x}$;

and $y_p'''(x) = (8Ax + 12A + 8B)e^{2x}$

d) The general solution of the linear nonhomogeneous differential equation $2y''' - 3y'' - 11y' + 6y = xe^{2x}$ is $y(x) = y_c(x) + y_p(x) =$ ___?___.

Question 5:

Find the general solution of the linear nonhomogeneous differential equation $4y^{(4)} - 4y''' - 47y'' + 114y' - 72y = x^2 + 2x + 5$

The linear homogeneous differential equation is $4y^{(4)} - 4y''' - 47y'' + 114y' - 72y = 0.$

The indicial equation is $4r^4 - 4r^3 - 47r^2 + 114r - 72 = 0.$

Note: $4r^4 - 4r^3 - 47r^2 + 11r - 72 = (r-2)(r+4)(2r-3)^2$.

a) The roots of the indicial equation $4r^4 - 4r^3 - 47r^2 + 11r - 72 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, $r_3 =$ ___?___, and $r_4 =$ ___?___.
Note: $r_3 = r_4$ is a repeated root.

b) The complementary solution $y_c(x)$ of $4y^{(4)} - 4y''' - 47y'' + 114y' - 72y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_p(x)$ for $g(x) = x^2 + 2x + 5$ is $y_p(x) =$ ___?___.
Hint: $y_p(x) = Ax^2 + Bx + C$; $y_p'(x) = 2Ax + B$; $y_p''(x) = 2A$; $y_p'''(x) = 0$; and $y^{(4)}(x) = 0.$

d) The general solution of the linear nonhomogeneous differential equation $4y^{(4)} - 4y''' - 47y'' + 114y' - 72y = x^2 + 2x + 5$ is
$y(x) = y_c(x) + y_p(x) =$ ___?___.

Question 5:

Find the general solution of the linear nonhomogeneous differential equation $4y^{(4)} - 4y''' - 47y'' + 11y' - 72y = x^2 + 2x + 5$

The linear homogeneous differential equation is $4y^{(4)} - 4y''' - 47y'' + 11y' - 72y = 0.$

The indicial equation is $4r^4 - 4r^3 - 47r^2 + 11r - 72 = 0.$

Note: $4r^4 - 4r^3 - 47r^2 + 11r - 72 = (r-2)(r+4)(2r-3)^2$.

a) The roots of the indicial equation $2r^3 - 3r^2 - 11r + 6 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, $r_3 =$ ___?___, and $r_4 =$ ___?___.
Note: $r_3 = r_4$ is a repeated root.

b) The complementary solution $y_c(x)$ of $4y^{(4)} - 4y''' - 47y'' + 11y' - 72y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_p(x)$ for $g(x) = x^2 + 2x + 5$ is $y_p(x) =$ ___?___.
Hint: $y_p(x) = Ax^2 + Bx + C$; $y_p'(x) = 2Ax + B$; $y_p''(x) = 2A$; $y_p'''(x) = 0$; and $y^{(4)}(x) = 0.$

d) The general solution of the linear nonhomogeneous differential equation $4y^{(4)} - 4y''' - 47y'' + 11y' - 72y = x^2 + 2x + 5$ is $y(x) = y_c(x) + y_p(x) =$ ___?___.

Question 6:

Find the general solution of the linear nonhomogeneous differential equation $2y''' + 3y'' - 41y' + 60y = x^2 + 3x - 1$.
Then find the particular solution with the initial conditions $y''(0) = 1$, $y'(0) = -1$, and $y(0) = 2$.

The linear homogeneous differential equation is $2y''' + 3y'' - 41y' + 60y = 0$.
The indicial equation is $2r^3 + 3r^2 - 41r + 60 = 0$.
Note: $2r^3 + 3r^2 - 41r + 60 = (r-3)(r+4)(2r-5)$.

a) The roots of the indicial equation $2r^3 + 3r^2 - 41r + 60 = 0$ are $r_1 =$ ___?___, $r_2 =$ ___?___, and $r_3 =$ ___?___.

b) The complementary solution $y_c(x)$ of $2y''' + 3y'' - 41y' + 60y = 0$ is $y_c(x) =$ ___?___.

c) The particular solution $y_p(x)$ for $g(x) = x^2 + 3x - 1$ is $y_p(x) =$ ___?___.
Hint: $y_p(x) = Ax^2 + Bx + C$; $y_p'(x) = 2Ax + B$; $y_p''(x) = 2A$; $y_p'''(x) = 0$; and $y^{(4)}(x) = 0$.

d) The general solution of the linear nonhomogeneous differential equation $2y''' + 3y'' - 41y' + 60y = x^2 + 3x - 1$ is
$y(x) = y_c(x) + y_p(x) =$ ___?___.

e) Using the result in part (d): $y'(x) =$ ___?___ and $y''(x) =$ ___?___.

f) The result in part (d) has the constants C_1, C_2, and C_3.

Use the initial conditions $y''(0) = 1$, $y'(0) = -1$, and $y(0) = 2$ to find C_1, C_2, and C_3.

$C_1 =$ ___?___ $C_2 =$ ___?___ and $C_3 =$ ___?___